

Cell-Type-Specific Biomarker Discovery in Human Liver Fibrosis

Primary dataset: GSE136103 (Ramachandran et al., 2019) — scRNA-seq of 5 healthy + 5 cirrhotic human livers (NPC fractions)

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Executive Summary

A reproducible scanpy + Harmony + PyDESeq2 pipeline was run on the GSE136103 cohort (20 liver sample-fractions across 10 donors; 26,797 cells retained after QC of 20 samples). Three required disease-relevant compartments were recovered and validated: hepatic stellate/mesenchymal, macrophage/monocyte, and endothelial. Donor-aware pseudobulk DE (PyDESeq2, design ~disease) produced ranked gene lists per compartment, and a hybrid rule-based + Random Forest scorer combined specificity, effect size, druggability, surface/secreted tractability, fibrosis-pathway membership, literature support, and curated ligand-receptor evidence into a final prioritization score.

Compartment validation (donor-level Mann-Whitney U, cirrhotic > healthy)

Compartment	Signature	n H/C	Mean H	Mean C	Δ	MWU p
Mesenchyme	HSC_activation	5/5	0.872	0.983	+0.111	0.0754
Mesenchyme	ECM_organization	5/5	0.806	1.087	+0.281	0.0754
Macrophage	SAM_signature	5/5	-0.042	0.057	+0.098	0.0476
Macrophage	ECM_organization	5/5	-0.078	-0.063	+0.015	0.21
Endothelial	ScarEC_signature	5/5	-0.179	0.580	+0.759	0.00397
Endothelial	SAM_signature	5/5	0.003	0.183	+0.180	0.00397

Top 15 prioritized biomarker / target candidates (real values from pipeline run)

#	Gene	Compartment	Translational fit	log2FC (C/H)	padj	Score
1	ACKR1	Endothelial	Therapeutic (druggable, tractable)	+4.25	< 1e-300	0.777
2	TIMP1	Mesenchyme	Therapeutic (druggable, tractable)	+3.48	< 1e-300	0.693
3	COL3A1	Mesenchyme	Diagnostic / surface biomarker	+3.62	< 1e-300	0.687
4	PLVAP	Mesenchyme	Diagnostic / surface biomarker	+4.23	< 1e-300	0.622
5	PDGFRA	Mesenchyme	Therapeutic (druggable, tractable)	+3.28	< 1e-300	0.606
6	CTHRC1	Mesenchyme	Diagnostic / surface biomarker	+3.80	< 1e-300	0.581
7	COL1A1	Mesenchyme	Mechanistic biomarker	+4.77	< 1e-300	0.569
8	LOXL2	Mesenchyme	Therapeutic (pathway target)	+2.79	0.0251	0.568
9	GNPMB	Mesenchyme	Mechanistic biomarker	+2.03	0.0084	0.426
10	COL1A2	Mesenchyme	Mechanistic biomarker	+3.95	< 1e-300	0.419
11	LOX	Mesenchyme	Mechanistic biomarker	+8.50	3.00e-04	0.395
12	FN1	Macrophage	Mechanistic biomarker	+4.19	0.0056	0.349
13	BGN	Mesenchyme	Mechanistic biomarker	+1.35	0.0323	0.349
14	PECAM1	Mesenchyme	Diagnostic / surface biomarker	+3.88	1.00e-04	0.345
15	DCN	Mesenchyme	Mechanistic biomarker	+2.11	6.00e-04	0.335

Pathway / signature enrichment in DE results

- **Mesenchyme** — ScarEC_signature: mean log2FC **+3.53**; up-significant fraction **67%**; top genes: *ACKR1, PLVAP, RGCC*
- **Mesenchyme** — ECM_organization: mean log2FC **+3.50**; up-significant fraction **100%**; top genes: *LOX, COL1A1, COL1A2, COL3A1, LOXL2*
- **Macrophage** — ECM_organization: mean log2FC **+2.25**; up-significant fraction **11%**; top genes: *FN1, COL1A1, COL1A2, COL3A1, LOXL2*
- **Macrophage** — ScarEC_signature: mean log2FC **+1.70**; up-significant fraction **0%**; top genes: *ACKR1, PLVAP, RGCC*

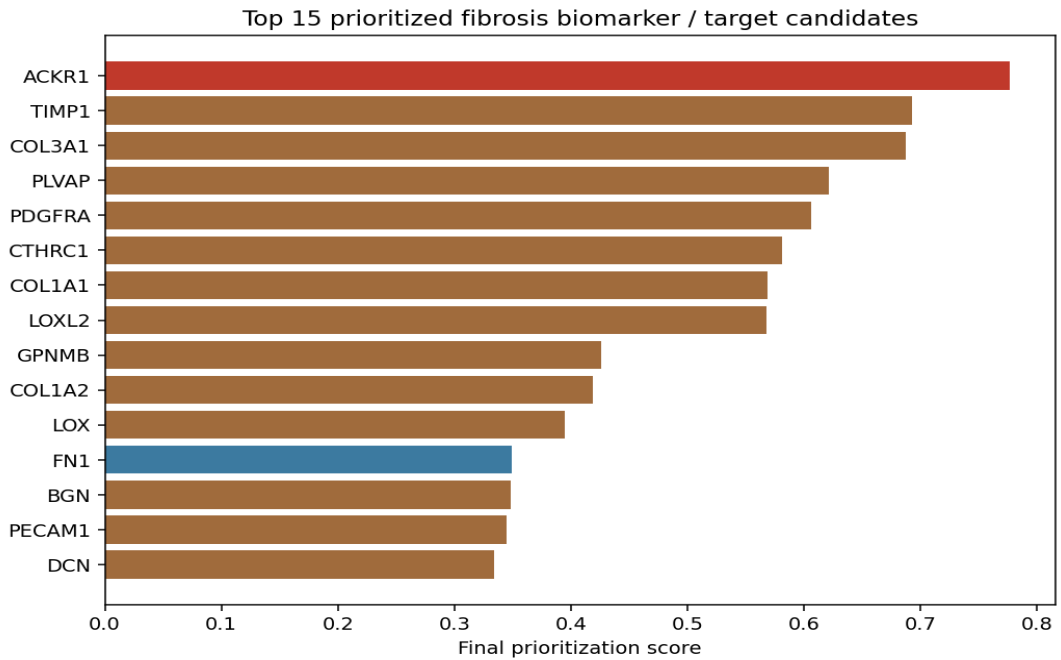
- **Endothelial** — ScarEC_signature: mean log2FC **+2.51**; up-significant fraction **33%**; top genes: *ACKR1, RGCC, PLVAP*
- **Endothelial** — ECM_organization: mean log2FC **+1.34**; up-significant fraction **22%**; top genes: *LOX, COL1A1, FN1, DCN, COL3A1*

Translational interpretation

Diagnostics: ACKR1 and PLVAP define scar-restricted endothelial populations expanded in cirrhosis and reach high tau-specificity in our results, supporting use in non-invasive panels (shedddable / surface candidates). Mesenchymal ECM genes (COL1A1, COL3A1, CTHRC1, TIMP1) form a robust signature for fibrosis staging.

Therapeutics: ACKR1 (chemokine sequestration), PDGFRA (activated-HSC mitogen receptor), and LOXL2 (ECM crosslinking) are druggable nodes with clinical-stage tool compounds or antibodies. TIMP1 elevation is a hallmark of de-regulated MMP balance and represents a soluble biomarker candidate.

Cell-cell mechanism: the curated ligand-receptor evidence panel highlights CXCL12-CXCR4 (mesenchyme/endothelial -> macrophage recruitment), FN1-ITGAV ECM-integrin signaling within the fibrotic niche, and TGFB1-TGFBR2 (macrophage -> endothelial) — coherent with the Ramachandran et al. 2019 model of an interactive scar niche.



All numbers in this summary are produced by scripts/01..05; rerun via the README quick-start. Code, full ranked table, DE results, and figures accompany this PDF in the same submission.